Study Report: Autonomous Surveillance Drone Path Planning

(Initial Study Report)

Prepared by:
Aman Shukla
Summer Research Intern
IIT Jodhpur
June 2026

1. Introduction

As part of the assigned task, I studied the concepts involved in autonomous surveillance drone path planning and the technologies required for autonomous navigation.

The study focused on GPS, IMU, Camera Systems, Sensor Fusion, Kalman Filter, Waypoint Navigation, Obstacle Avoidance, and Path Planning Algorithms (A* and RRT).

The objective of this study is to understand how a surveillance drone can determine and follow a safe path from a start location to a target location while avoiding obstacles.

2. Main Components of an Autonomous Surveillance Drone

Hardware Components

- Flight Controller
- Raspberry Pi
- GPS Module
- IMU Sensor
- Camera Module
- Battery System
- Communication Module
- Motors and ESCs

Software Components

- Python Programming
- Path Planning Algorithms

- Sensor Fusion Algorithms
- Navigation Logic

3. GPS (Global Positioning System)

GPS provides the geographical position of the drone and enables waypoint-based navigation.

Functions:

- Position estimation
- Route tracking
- Waypoint navigation
- Return-to-home functionality

Common Modules:

- u-blox NEO-6M
- u-blox M8N

4. IMU (Inertial Measurement Unit)

The IMU provides information about orientation and motion.

Components:

- Accelerometer
- Gyroscope
- Magnetometer

Functions:

- Roll estimation
- Pitch estimation
- Yaw estimation
- Motion sensing

Common Sensors:

- MPU6050
- MPU9250

5. Camera System

The camera is used for surveillance and image acquisition.

Applications:

- Monitoring
- Object detection
- Video recording
- Visual navigation

Possible Choice:

- Raspberry Pi Camera Module

6. Sensor Fusion

GPS and IMU data individually contain errors. Sensor fusion combines information from multiple sensors to obtain a more accurate estimate of position and orientation.

Benefits:

- Improved accuracy
- Better navigation performance
- Increased reliability

7. Kalman Filter

The Kalman Filter is commonly used for sensor fusion.

Purpose:

- Reduce sensor noise
- Improve localization accuracy
- Estimate drone state

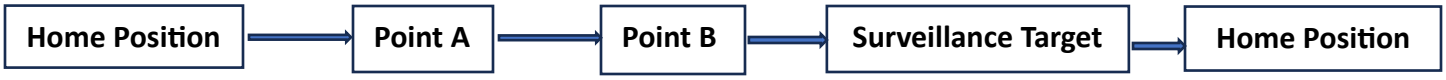
Concept:



8. Waypoint Navigation

Waypoints are predefined coordinates that the drone follows during a mission.

Example Mission:



9. Path Planning

Path planning determines a safe and efficient route between the drone's current location and the surveillance target.

Objectives:

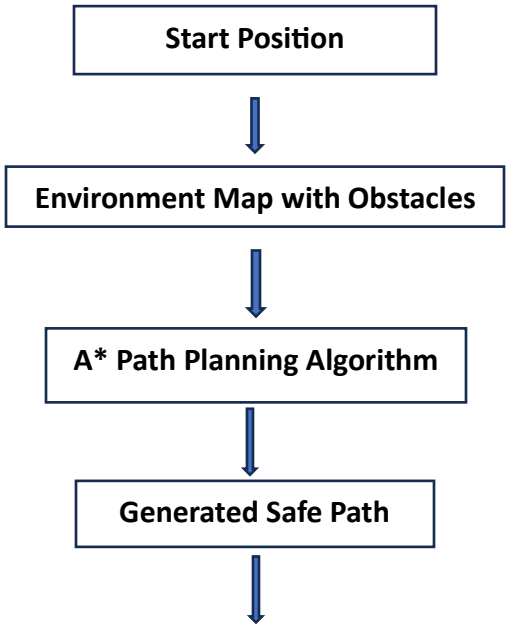
- Shortest path generation
- Safe navigation
- Obstacle avoidance
- Efficient battery utilization

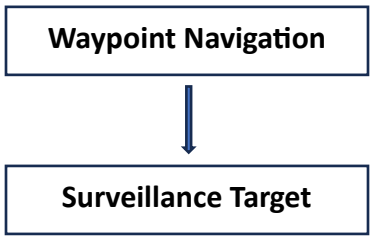
For the initial implementation, I propose using the A* (A-Star) Path Planning Algorithm.

Reason for Selection:

- Easy to implement and validate in Python
- Generates an optimal path in known environments
- Computationally efficient
- Suitable for waypoint-based navigation
- Can be deployed later on Raspberry Pi

Initial Workflow:





10. Obstacle Avoidance

Obstacle avoidance prevents collisions during flight.

Possible Sensors:

- Camera
- Ultrasonic Sensor
- LiDAR

Examples of Obstacles:

- Buildings
- Trees
- Poles
- Restricted Zones

11. A* Path Planning Algorithm

A* is a graph-based path planning algorithm that finds an optimal path between a start node and a goal node.

Cost Function:

$f(n) = g(n) + h(n)$

Where:

- g(n): Cost from start to current node
- h(n): Estimated cost from current node to goal

Advantages:

- Optimal path generation
- Efficient implementation
- Suitable for embedded platforms

- Easy simulation using Python

Reason for Selection:

A* is selected as the initial implementation algorithm because of its simplicity and effectiveness for known environments.

12. RRT (Rapidly Exploring Random Tree)

RRT is a sampling-based path planning algorithm.

Advantages:

- Suitable for complex environments
- Handles large search spaces

Limitations:

- More complex implementation
- Generated path may not be optimal

13. Proposed Implementation Approach

Phase 1:

- Study GPS, IMU, Sensor Fusion, and Path Planning concepts
- Implement A* Algorithm in Python
- Test path generation between a start position and surveillance target
- Validate obstacle avoidance in a simulated environment

Phase 2:

Validate path planning through simulation

- Analyze generated routes and performance

Phase 3

- Explore Raspberry Pi deployment
- Execute path planning algorithm on Raspberry Pi

Phase 4

- Integrate with drone hardware and sensors
- Study waypoint-based navigation on an actual platform

14. Additional Study

Along with the path planning study, I installed:

- Keil uVision IDE
- Python Development Environment

I also studied the STM32F4 Digital Systems Laboratory Manual.

Topics covered:

- STM32F4 Architecture Overview
- Embedded System Development Flow
- GPIO Basics
- Keil uVision Development Environment

15. Current Progress

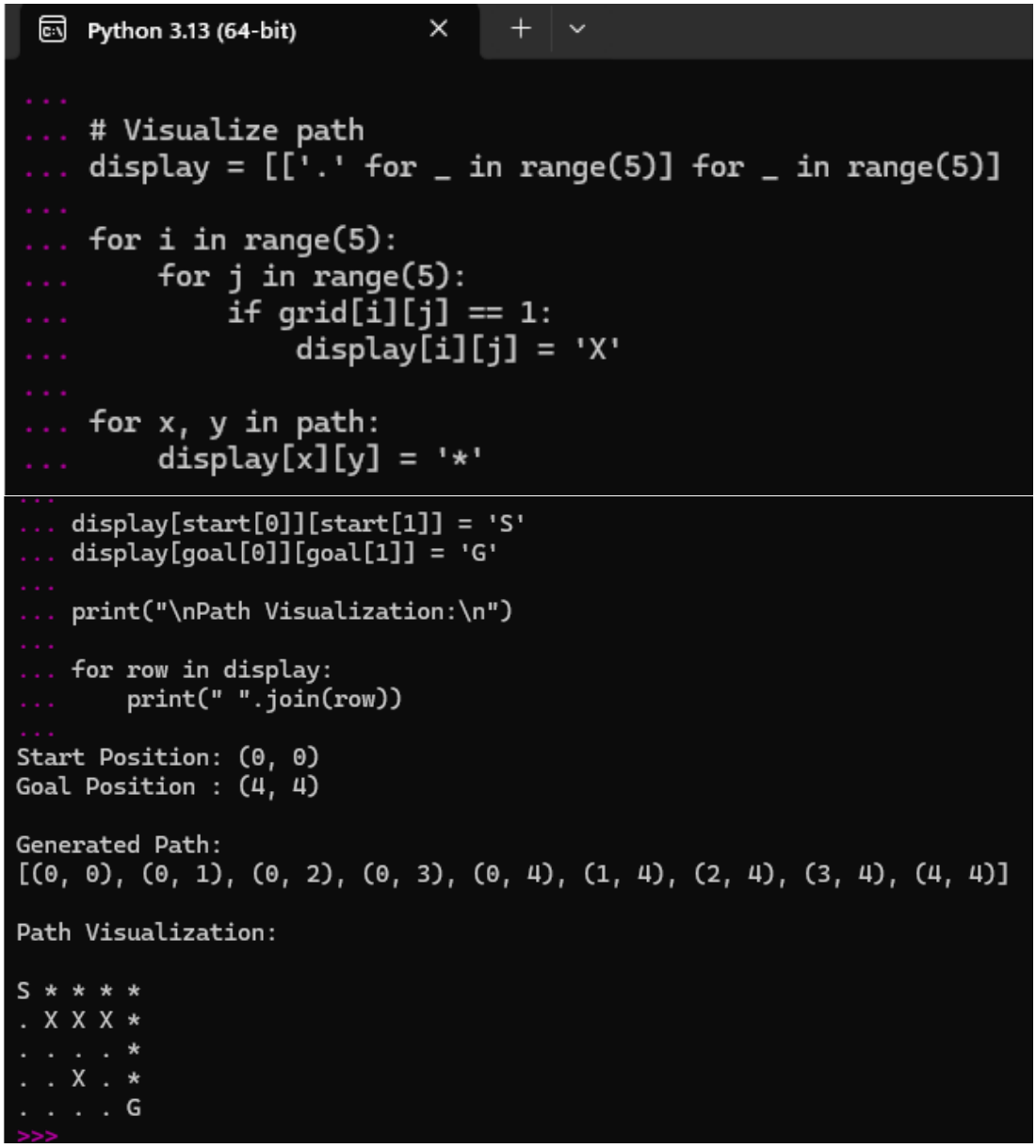
Completed:

- STM32F4 Manual Study
- Installation of Keil uVision
- Installation of Python
- Study of Autonomous Drone Fundamentals
- Study of GPS, IMU, Camera, Sensor Fusion, and Kalman Filter
- Study of A* and RRT Algorithms

Currently Working On:

- Implementation of A* Path Planning Algorithm in Python
- Testing path generation in a simulated environment

16. Initial Python Implementation



A basic implementation of the A* path planning algorithm was developed and tested in Python.

Test Parameters:

- Start Position: (0,0)
- Goal Position: (4,4)
- Grid-based Environment
- Obstacles placed within the map

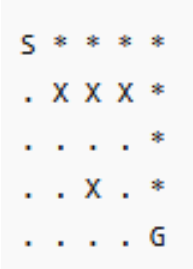
Output:

The algorithm successfully generated a collision-free path from the start position to the target position while avoiding obstacles.

Generated Path:

(0,0) → (0,1) → (0,2) → (0,3) → (0,4) → (1,4) → (2,4) → (3,4) → (4,4)

Path Visualization:



Where:

- S = Start Position
- G = Goal Position
- X = Obstacle
- * = Generated Path

17. Conclusion

The study provided an understanding of autonomous surveillance drone architecture, navigation sensors, sensor fusion techniques, waypoint navigation, and path planning algorithms.

Based on the initial study, A* appears to be a suitable algorithm for the first implementation phase due to its simplicity and effectiveness in generating safe paths in known environments.